

A counterexample to injectivity of the automorphism-to-homeomorphism map

arXiv:2604.10285, Remark 6.16

June 14, 2026

Source metadata

The source paper is M.H.M. Rashid, *Operator Algebras of Bourgain Delbaen Spaces: Realization, Rigidity, and Ideal Structure*, arXiv:2604.10285. Remark 6.16, on page 23 of the source PDF, asks about the automorphism group of $\mathcal{L}(\mathfrak{X}_{C(K)})$ and the natural map from automorphisms to homeomorphisms of K .

Source question

The relevant source remark is reproduced below.

applies more generally to any class of Banach spaces satisfying:

- $\mathcal{Cal}(X) \cong C(K)$ for some compact metric space K , with the isomorphism being canonical and preserving the algebraic structure,
- The compact operators form a characteristic ideal (e.g., the unique minimal closed two-sided ideal),
- The diagonal-plus-compact decomposition holds, and the diagonal function φ_T is well-defined and continuous.

Many variations of the Bourgain-Delbaen construction satisfy these properties, so the rigidity phenomenon is robust.

Remark 6.16 (Open Questions). *Theorem 6.7 raises natural questions about the automorphism group of $\mathcal{L}(\mathfrak{X}_{C(K)})$. Does every automorphism come from a homeomorphism of K ? The naturality condition suggests that the map $\text{Aut}(\mathcal{L}(\mathfrak{X}_{C(K)})) \rightarrow \text{Homeo}(K)$ given by $\Phi \mapsto h$ might be a surjective homomorphism. Is it injective? That is, can there exist non-inner automorphisms that induce the identity on K ? This question touches on the structure of the automorphism group and remains open for further investigation.*

Theorem 6.7 establishes a profound connection between the topology of a compact metric space and the algebraic structure of the bounded operators on an associated Banach space. It demonstrates that the Bourgain-Delbaen construction creates spaces whose operator algebras are so rigid that they completely remember the geometry that went into their construction. This represents a significant advance in our understanding of how geometric data can be encoded in operator algebras, and opens new avenues for the classification of Banach spaces through their operator algebras.

23

In particular, the source asks whether the natural map

$$\text{Aut}(\mathcal{L}(\mathfrak{X}_{C(K)})) \longrightarrow \text{Homeo}(K), \quad \Phi \longmapsto h,$$

obtained from the induced action on $\mathcal{Cal}(\mathfrak{X}_{C(K)}) \cong C(K)$, might be injective.

Proof intuition

The kernel of this map is already large at the elementary level. If an invertible operator U is a compact perturbation of the identity, then conjugation by U is invisible in the Calkin algebra: modulo compact operators, U and U^{-1} are both just the identity. Nevertheless, conjugation by such a U need not be the identity automorphism of the full operator algebra. Taking $U = I + P$ for a nonzero finite-rank projection P gives an explicit witness.

Counterexample

Theorem. *Let X be any Banach space with $\dim X \geq 2$. The quotient-induced map from automorphisms of $\mathcal{L}(X)$ to automorphisms of the Calkin algebra is not injective. Consequently, for every space $\mathfrak{X}_{C(K)}$ in the source paper, the stated map*

$$\text{Aut}(\mathcal{L}(\mathfrak{X}_{C(K)})) \longrightarrow \text{Homeo}(K)$$

is not injective.

Proof. Choose $x \in X$ and $f \in X^*$ with $f(x) = 1$, and set

$$Pz = f(z)x \quad (z \in X).$$

Then P is a rank-one projection. The operator $U = I + P$ is invertible, with inverse $U^{-1} = I - \frac{1}{2}P$. Hence

$$\Phi(T) = U T U^{-1} \quad (T \in \mathcal{L}(X))$$

defines an automorphism of the Banach algebra $\mathcal{L}(X)$.

This automorphism induces the identity on the Calkin algebra. Indeed, $U - I = P$ is finite-rank, hence compact, so in the quotient map $q : \mathcal{L}(X) \rightarrow \text{Cal}(X) = \mathcal{L}(X)/\mathcal{K}(X)$ we have $q(U) = q(I)$ and $q(U^{-1}) = q(I)$. Therefore, for every $T \in \mathcal{L}(X)$,

$$q(\Phi(T)) = q(U)q(T)q(U^{-1}) = q(T).$$

It remains only to check that Φ is not the identity automorphism. Pick $0 \neq y \in \ker f$, and define the rank-one operator

$$Tz = f(z)y \quad (z \in X).$$

Then $PT = 0$, while $TP = T$. Hence

$$\Phi(T) = (I + P)T\left(I - \frac{1}{2}P\right) = (I + P)\left(T - \frac{1}{2}TP\right) = (I + P)\frac{1}{2}T = \frac{1}{2}T \neq T.$$

Thus $\Phi \neq \text{id}_{\mathcal{L}(X)}$, but Φ has the same induced Calkin-algebra action as the identity automorphism.

For $X = \mathfrak{X}_{C(K)}$, the source paper identifies $\text{Cal}(X)$ with $C(K)$. Since Φ induces the identity on $\text{Cal}(X)$, it induces the identity automorphism of $C(K)$, and hence the identity homeomorphism of K under the Banach-Stone/Gelfand identification used in the source paper. Therefore the automorphisms $\text{id}_{\mathcal{L}(X)}$ and Φ are distinct but have the same image in $\text{Homeo}(K)$. The map is not injective. $\square$

Remark. *The automorphism constructed above is inner. Thus the argument answers the literal injectivity question in Remark 6.16, but it does not decide whether there is an outer automorphism in the kernel. If the intended problem is the induced map after quotienting by inner automorphisms, that is a strictly stronger formulation and is not settled here.*

Verification and novelty check

No computational verification is needed; the proof is the finite-rank calculation above. The open-question crop is from page 23 of the source PDF. Local duplicate checks over the run registry, solution index, attempt index, proof-gap index, and active claims found nearby Calkin algebra packets but no packet for arXiv:2604.10285 or this automorphism-kernel question.

A bounded web search on 2026-06-14 for exact and close phrases such as

" $\text{Aut}(\mathcal{L}(\mathcal{K}))$ " " $\text{Homeo}(K)$ "

and

"Operator Algebras of Bourgain" "Is it injective"

found no prior answer to the exact Remark 6.16 injectivity question. Novelty confidence is therefore moderate: the obstruction is elementary, so it may be folklore, but it was not found in the bounded search.

References

- [1] M.H.M. Rashid, *Operator Algebras of Bourgain Delbaen Spaces: Realization, Rigidity, and Ideal Structure*, arXiv:2604.10285, 2026.

Human Review: automorphism-map noninjectivity

Original automatic proof: `solution_packet.pdf`

Checked date: 14 June 2026

Status: Reviewed for survey inclusion, but not independently checked in detail by the human reviewer; excluded for the time being.

Verdict: Exclude from the verified successful examples at this stage.

Human Comments

The proposed counterexample is an elementary observation about the literal formulation of the question. It argues as follows. If X is a Banach space and P is a nonzero finite-rank projection, then

$$U = I + P$$

is invertible, and conjugation

$$\text{Ad}_U(T) = UTU^{-1}$$

defines an automorphism of $\mathcal{L}(X)$. Since $U - I = P$ is compact, this automorphism induces the identity map on the Calkin algebra

$$\text{Cal}(X) = \mathcal{L}(X)/\mathcal{K}(X).$$

The packet further argues that Ad_U need not be the identity automorphism of $\mathcal{L}(X)$. Thus, if this elementary calculation is accepted, then the literal map from automorphisms of $\mathcal{L}(X)$ to the induced automorphisms of the Calkin algebra would not be injective.

However, I do not think this should be included as a successful project example for the time being. The source manuscript itself raises several reliability concerns. It appears to claim a large number of substantial results at once, including realization, rigidity, ideal classification, and new reflexive Calkin-algebra examples. Some parts of the manuscript also look internally suspicious. For example, the running header refers to “Strongly Singular Kirchhoff-Type Equations”, which is unrelated to the paper title and content. More substantively, the paper appears to pass from finite direct sums of copies of $C(K)$ to matrix algebras $M_m(C(K))$ by using a linear identification, even though these are not isomorphic as Banach algebras for $m > 1$ since the former is commutative and the latter is not. There is also a claim involving ℓ_2 with pointwise multiplication as a Calkin algebra, which is problematic since such an algebra is not unital.

There is a second issue with the formulation of the relevant open question. The remark asks about injectivity of a natural map to $\text{Homeo}(K)$, but it also phrases the issue in terms of possible non-inner automorphisms in the kernel. The candidate counterexample only supplies an inner automorphism, coming from conjugation by an invertible compact perturbation of the identity. Therefore it answers the literal wording, but probably not the more meaningful intended question if inner automorphisms were supposed to be ignored.

Short Version

The finite-rank conjugation argument is a possible elementary objection to the literal wording: inner automorphisms implemented by invertible compact perturbations of the identity should be invisible in the Calkin algebra. But the argument has not been independently checked in detail by the human reviewer, the source paper appears unreliable enough, and the intended question may differ enough from the literal wording that this packet should be excluded for now.

Literature and Follow-Up

This example should not be used as evidence of mathematical progress unless the source manuscript is independently validated or the authors clarify that the literal inner-automorphism obstruction is genuinely relevant to their intended question. If the intended problem concerns outer automorphisms or automorphisms modulo inner automorphisms, then the present packet does not address it.

Review Outcome

Exclude this packet from the verified successful examples for the time being. It may be retained as a cautionary case: the automatic system found a possible elementary objection to the literal wording, but the argument has not been independently checked in detail by the human reviewer, and the surrounding source and likely intended question make it unsuitable for inclusion as a positive mathematical-discovery example.